

SALES DASHBOARD

Data Visualisation – Tableau Workbook

Technical Documentation

Table of Contents

1. Overview	3
2. Dashboard: Sales Dashboard	4
3. Data Sources	5
4. Data Model – Field Reference	6
5. Calculated Fields.....	8
6. Worksheets.....	10
8. Color Palette.....	13
9. Setup & Deployment Notes	Error! Bookmark not defined.

1. Overview

This document provides a comprehensive technical description of the Sales Dashboard Tableau workbook. The workbook was built in Tableau Desktop 2026.1 and is designed to give sales analysts and business stakeholders a single interactive view of key performance across sales, profit, and order quantity segmented by time, geography, customer segment, and product category.

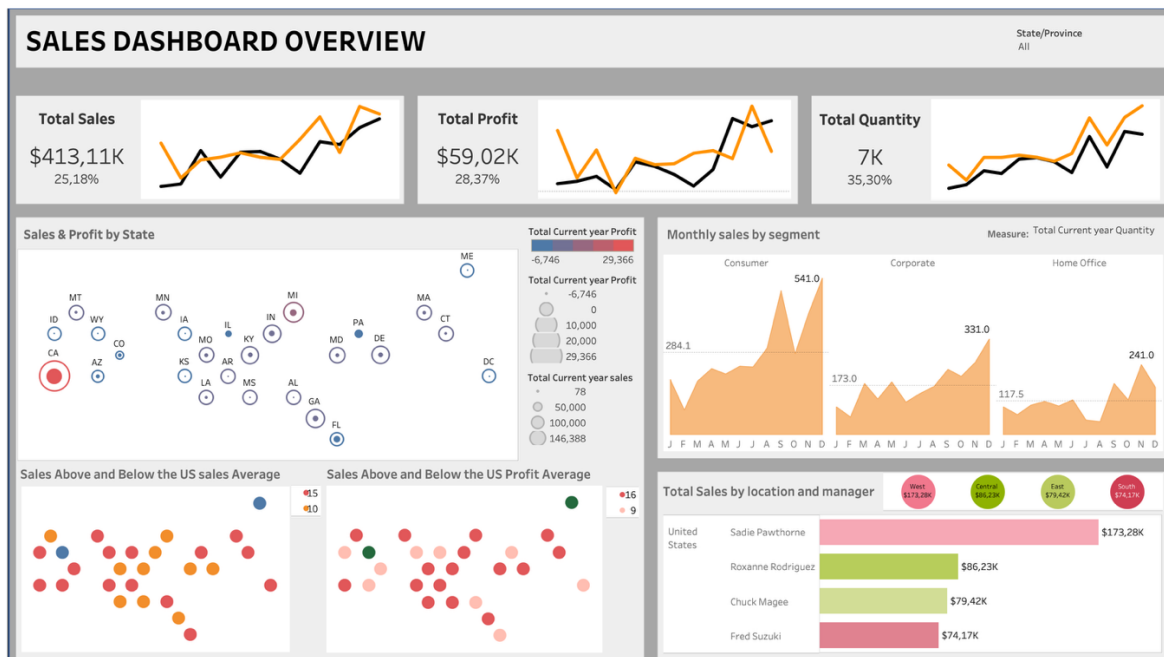
The dashboard supports year-over-year comparison through calculated KPI measures and includes geographic detail at the US state level via both standard maps and a hexmap layout. A dynamic parameter allows users to toggle the primary measure between Sales, Profit, and Quantity without switching sheets.

2. Dashboard: Sales Dashboard

The workbook contains a single published dashboard named Sales Dashboard. It is configured to open maximized and assembles the following 11 worksheets into a cohesive layout:

- Sales KPI + Sales Sparkline (KPI banner – top row)
- Profit KPI + Profit Sparkline (KPI banner – top row)
- Quantity KPI + Quantity Sparkline (KPI banner – top row)
- Sales and Profit by State (geographic map, center)
- AVG Sales by State / AVG Profit by State (bar comparison charts)
- Monthly Sales by Segment (time-series by segment, lower section)
- Total Sales by Location and Manager (hexmap, lower right)
- Region Wise Sales (regional summary bar chart)

Eleven auto-generated filter actions (Filter 1 through Filter 11) link the sheets together so that clicking a mark on any sheet cross-filters the rest of the dashboard.



3. Data Sources

The workbook connects to three federated data sources, all sourced from the local file Sales Data.xls.

Data Source	Description	Key Fields Included
Sales Data.xls – Orders	Primary transactional order data (CSV import). Contains 10,000 rows of order records.	Row ID, Order ID, Order Date, Ship Date, Ship Mode

Note: The hexmap.xlsx reference file provides Abbreviation, Column, and Row coordinates used to render the US state hexmap visualization in the ‘Total Sales by Location and Manager’ worksheet.

4. Data Model – Field Reference

The following table documents all 19 fields available from the core Orders dataset.

Field Name	Data Type	Description
Row ID	Integer	Unique row identifier
Order ID	String	Unique order identifier
Order Date	Date	Date the order was placed
Ship Date	Date	Date the order was shipped
Ship Mode	String	Shipping method (Standard, Second Class, etc.)
Customer ID / Name	String	Customer identifier and full name
Segment	String	Customer segment: Consumer, Corporate, Home Office
Country / Region	String	Country of the order
City	String	City of delivery
State / Province	String	US state or province
Postal Code	Integer	Delivery postal code
Region	String	Sales region: East, West, Central, South
Product ID / Name	String	Product identifier and full product name
Category	String	Product category: Furniture, Office Supplies, Technology
Sub-Category	String	Product sub-category (17 sub-categories)

Field Name	Data Type	Description
Sales	Decimal	Revenue value of the order line
Quantity	Integer	Number of units ordered
Discount	Decimal	Discount rate applied (0.0 – 1.0)
Profit	Decimal	Net profit on the order line

5. Calculated Fields

The workbook defines 16 calculated fields to power the KPI cards, sparklines, and color-coded comparison visuals. These fields are all housed in the Orders+ (Sales Data) data source.

Calculated Field	Logic / Purpose
Total Current Year Sales	SUM of Sales for the selected/current year
Total Last Year Sales	SUM of Sales for the prior year
Year on Year Sales	Current Year Sales – Last Year Sales (absolute variance)
Total Current Year Profit	SUM of Profit for the current year
Total Last Year Profit	SUM of Profit for the prior year
Year on Year Profit	Current Year Profit – Last Year Profit
Total Current Year Quantity	SUM of Quantity for the current year
Total Last Year Quantity	SUM of Quantity for the prior year
Year on Year Quantity	Current Year Quantity – Last Year Quantity
AVG Sales Overall	Overall average sales across all states
Avg Sales State Wise	Average sales per state
Sales Average Colour	Conditional string dimension: 'Above Average' / 'Below Average' for color encoding
AVG Profit Overall	Overall average profit across all states
Avg Profit State Wise	Average profit per state

Calculated Field	Logic / Purpose
Profit Average Colour	Conditional string dimension for above/below average profit color encoding
Dynamic Measure	Parameter-driven measure swapping between Sales, Profit, and Quantity

The 'Select Measure' parameter (Parameter 1) drives the Dynamic Measure calculated field and is pre-set to 'Total Current Year Quantity'. Users can change it to switch the primary metric displayed across compatible sheets.

6. Worksheets

The workbook contains 14 worksheets plus one backend calculation sheet. The table below summarises the purpose of each.

Worksheet Name	Purpose & Visualization Type
Sales KPI	KPI card showing current year total sales vs last year with year-on-year % change.
Profit KPI	KPI card for current year profit vs last year with year-on-year variance.
Quantity KPI	KPI card for units sold in current year vs prior year with trend indicator.
Sales Sparkline	Line sparkline chart showing monthly sales trend for the current year.
Profit Sparkline	Line sparkline chart showing monthly profit trend for the current year.
Quantity Sparkline	Line sparkline chart showing monthly quantity trend for the current year.
Sales and Profit by State	Filled map or bar chart combining state-level sales and profit data.
AVG Sales by State	Bar/dot chart comparing state-level average sales vs overall average, color-coded above/below average.
AVG Profit by State	Bar/dot chart comparing state-level average profit vs overall average, color-coded above/below average.
AVG Sales by State Count	Summary count of states performing above vs below the average sales benchmark.
AVG Profit by State Count	Summary count of states performing above vs below the average profit benchmark.

Worksheet Name	Purpose & Visualization Type
Monthly Sales by Segment	Bar or line chart of monthly sales broken down by customer segment (Consumer, Corporate, Home Office).
Total Sales by Location & Manager	Hexmap or geographic view of sales by location linked to regional sales manager data.
Region Wise Sales	Regional breakdown (East, West, Central, South) of total sales performance.
Sales KPI Calculations	Backend calculation sheet driving the KPI measures and year-on-year comparisons.

7. Interactivity & Actions

The dashboard uses 11 auto-generated filter actions to enable cross-sheet filtering:

- Filter 1 – Sales KPI → all dependent sheets
- Filter 2 – Sales Sparkline → all dependent sheets
- Filter 3 – Sales and Profit by State → AVG Sales / Profit sheets
- Filter 4 – AVG Sales by State → AVG Profit by State (and counts)
- Filter 5 – AVG Profit by State → reciprocal filter back
- Filters 6–11 – Additional cross-filters for Monthly Segment, Region, Hexmap sheets

All actions trigger on selection and are scoped to the Sales Dashboard. When a user selects a state on the map, all KPI, sparkline, and bar charts update to reflect that state's data.

The 'Select Measure' parameter adds a second layer of interactivity, allowing the user to toggle the dynamic metric across relevant visualizations without navigating away from the dashboard.

8. Color Palette

The workbook uses a custom two-color sequential palette named 'Blue-Red':

- Blue (#4E79A7) – represents above-average or positive performance
- Red (#E15759) – represents below-average or negative performance

This palette is applied consistently across the Sales Average Colour and Profit Average Colour dimensions to give users an immediate visual cue on state-level performance relative to the national average.